HOM OUT OF A FINITELY PRESENTED MODULE COMMUTES WITH FILTERED COLIMITS ON A QCQS STACK

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Allowed standard prerequisites (editorial).

- Finitely presented modules and filtered colimits. Modules on Sites, Lemma sites-modules-lemma-finite-presentation-quasi-compact-colimit. Module-level statement being instantiated.
- Directed colimits of global sections, hypotheses. Sites, Lemma sites-lemma-directed-colimits-global-sections (4) with Remark sites-remark-stronger-conditions. Provides the conditions (2)(a)-(c) verified.
- QCoh on an algebraic stack. Sheaves on Stacks, Lemma stacks-sheaves-lemma-quasi-coherent-algebraic-stack. Colimit commutation of the inclusion; fp modules are qc.

Source reference. Original reference 0GN0: sources/stacks-chow/tags.txt, source lines 20775–20775. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GMR: sources/stacks-chow/tags.txt, source lines 20766–20766. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GMS: sources/stacks-chow/tags.txt, source lines 20767–20767. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04YC: sources/stacks-chow/tags.txt, source lines 6425–6425. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 021P: sources/stacks-chow/tags.txt, source lines 2678–2678. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 075S: sources/stacks-chow/tags.txt, source lines 9141–9141. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WV: sources/stacks-chow/tags.txt, source lines 8829–8829. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{X}$ be a quasi-compact and quasi-separated algebraic stack. Let I be a directed set and let $(\mathcal{F}_i, \varphi_{ii'})$ be a system over I of $\mathcal{O}_\mathcal{X}$-modules. Let $\mathcal{G}$ be an $\mathcal{O}_\mathcal{X}$-module of finite presentation. Then we have*

$$\operatorname{colim}_i \operatorname{Hom}_\mathcal{X}(\mathcal{G}, \mathcal{F}_i) = \operatorname{Hom}_\mathcal{X}(\mathcal{G}, \operatorname{colim}_i \mathcal{F}_i).$$

In particular, $\operatorname{Hom}_\mathcal{X}(\mathcal{G}, -)$ commutes with filtered colimits in $QCoh(\mathcal{O}_\mathcal{X})$.

Proof. The displayed equality is a special case of Modules on Sites, Lemma 0GN0. In order to apply it, we need to check the hypotheses of Sites, Lemma 0GMR part (4) for the site $\mathcal{X}_{fppf}$. In order to do this, we will check hypotheses (2)(a), (2)(b), (2)(c) of Sites, Remark 0GMS. Namely, let $\mathcal{B} \subset \operatorname{Ob}(\mathcal{X}_{fppf})$ be the set of objects lying over affine schemes. In other words, an element of $\mathcal{B}$ is a morphism $x : U \to \mathcal{X}$ with U affine. We check each of the conditions (2)(a), (2)(b), and (2)(c) of the remark in turn:

(1) Since $\mathcal{X}$ is quasi-compact, there exists a surjective and smooth morphism $x : U \to \mathcal{X}$ with U affine (Properties of Stacks, Lemma 04YC). Then $h_x^\# \to *$ is a surjective map of sheaves on $\mathcal{X}_{fppf}$.
(2) Since coverings in $\mathcal{X}_{fppf}$ are fppf coverings, we see that every covering of $U \in \mathcal{B}$ is refined by a finite affine fppf covering, see Topologies, Lemma 021P.
(3) Let $x : U \to \mathcal{X}$ and $x' : U' \to \mathcal{X}$ be in $\mathcal{B}$. The product $h_x^\# \times h_{x'}^\#$ in $Sh(\mathcal{X}_{fppf})$ is equal to the sheaf on $\mathcal{X}_{fppf}$ determined by the algebraic space $W = U \times_{x,\mathcal{X},x'} U'$ over $\mathcal{X}$: for an object $y : V \to \mathcal{X}$ of $\mathcal{X}_{fppf}$ we have $(h_x^\# \times h_{x'}^\#)(y) = \{f : V \to W \mid y = x \circ \mathrm{pr}_1 \circ f = x' \circ \mathrm{pr}_2 \circ f\}$. The algebraic space W is quasi-compact because $\mathcal{X}$ is quasi-separated, see Morphisms of Stacks, Lemma 075S for example. Hence we can choose an affine scheme U'' and a surjective étale morphism $U'' \to W$. Denote $x'' : U'' \to \mathcal{X}$ the composition of $U'' \to W$ and $W \to \mathcal{X}$. Then $h_{x''}^\# \to h_x^\# \times h_{x'}^\#$ is surjective as desired.

For the final statement, observe that the inclusion functor $QCoh(\mathcal{O}_X) \to Mod(\mathcal{O}_X)$ commutes with colimits and that finitely presented modules are quasi-coherent. See Sheaves on Stacks, Lemma 06WV. □